$$mx'' = -kx + \frac{q^2V}{8\pi^2\varepsilon_0 z^5} + 5\frac{q^2V}{8\pi^2\varepsilon_0 z^6}x \tag{25}$$

.

$$k\Delta x = \frac{q^2V}{8\pi^2\varepsilon_0 z^5} \quad \Rightarrow \quad \Delta x = \frac{q^2V}{8\pi^2\varepsilon_0 z^5 k} \tag{26}$$

2.2.2 (24)

$$\omega = \sqrt{\frac{k}{m} - \frac{5q^2V}{8\pi^2\varepsilon_0 z^6 m}} = \sqrt{\omega_0^2 - \frac{5q^2V}{8\pi^2\varepsilon_0 z^6 m}} \approx \omega_0\left(1 - \frac{5q^2V}{16\pi^2\varepsilon_0 z^6 k}\right) \tag{27}$$

$$\frac{\Delta\omega}{\omega_0} = -\frac{5q^2V}{16\pi^2\varepsilon_0 z^6 k} \tag{28}$$

2.2.3 σ .

$(\dot{\sigma} = j = E/\rho).$

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(z-x)^2} \approx \frac{q}{4\pi\varepsilon_0 z^2}\left(1 + 2\frac{x}{z}\right)$$

$$E_2 = -\frac{\sigma}{\varepsilon_0}.$$

$E_1 + E_2 = \rho j$

$$\frac{q}{4\pi\varepsilon_0 z^2}\left(1 + 2\frac{x}{z}\right) - \frac{\sigma}{\varepsilon_0} = \rho j$$

$\dot{\sigma} = j$ $p = \sigma S h = \sigma V$ $(V = \pi R^2 h)$

$$\varepsilon_0\rho\dot{p} + p = \frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right).$$

2.2.4 $R = \rho\frac{h}{S}$, $C = \frac{\varepsilon_0 S}{h}$, $\tau = RC = \varepsilon_0\rho.$

2.2.5 $\frac{2\pi}{\omega} \gg \varepsilon_0\rho$ $\varepsilon_0\rho\omega \ll 1$..

2.2.6 $\varepsilon_0\rho\dot{p}$. p
$\varepsilon_0\rho\omega$, .

$$\varepsilon_0\rho\dot{p} + p = \frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right)$$

$$p = \frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right) \qquad \dot{p} = \frac{qV}{2\pi z^3}v$$

$\dot{p}$.

$$p = \frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right) - \varepsilon_0\rho\dot{p} = \frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right) - \varepsilon_0\rho\frac{qV}{2\pi z^3}v$$

.

2.2.7

$$F = \frac{p}{2\pi\varepsilon_0(z-x)^3}q = \frac{q}{2\pi\varepsilon_0 z^3}\left(1 + 3\frac{x}{z}\right)\left[\frac{qV}{4\pi z^2}\left(1 + 2\frac{x}{z}\right) - \varepsilon_0\rho\frac{qV}{2\pi z^3}v\right] =$$

$$= \frac{q^2R^2h}{8\pi\varepsilon_0 z^6}[(z + 5x) - 2\varepsilon_0\rho v]$$